

Miaomiao Dai

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EDUCATION

University of Michigan

B.S.E. in Computer Science (SJTU–UMich dual-degree sequence) · GPA 3.8/4.0

Aug 2024 – May 2026

Ann Arbor, MI

- Coursework: Robot Learning (A+), Algorithmic Robotics (A), Machine Learning (A), Computer Vision (A), Data Structures & Algorithms (A)

Shanghai Jiao Tong University

B.E. in Mechanical Engineering, Global College · GPA 3.6/4.0

Sep 2022 – Aug 2026

Shanghai, China

PUBLICATIONS

- [1] **Miaomiao Dai**, Zhongqiang Ren. “HOP: Fast Differential Dynamic Programming for Horizon-Optimal Trajectory Planning.” *Robotics: Science and Systems (RSS)*, 2026.
- [2] **Miaomiao Dai**, Zhongqiang Ren. “AL-HOP: Augmented-Lagrangian Horizon-Optimal Planning for Fast Constrained Trajectory Optimization.” Under review, 2026.

RESEARCH EXPERIENCE

HOP: Horizon-Optimal Trajectory Planning

Jul 2025 – Feb 2026

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · RSS 2026 talk, undergraduate first author

- Standard trajectory optimizers fix the task duration in advance and re-solve for every candidate; HOP selects the optimal horizon automatically within a single solve.
- Proposed and proved a **closed-form theory** that scores every candidate horizon in one backward sweep, cutting horizon search from $O(N^2)$ to $O(N)$ — up to **40×** faster than brute force, with identical selected horizon and cost.
- Designed and iterated the experimental evaluation across cartpole, Segway, and quadrotor systems.

AL-HOP: Constrained Horizon-Optimal Planning

Mar 2026 – present

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · first-author manuscript under review

- Extends HOP to hard state–control constraints (obstacles, actuator limits) with an augmented-Lagrangian formulation that keeps the one-sweep horizon prediction.
- Produced solutions valid under the full nonlinear dynamics on **100/100** benchmark problems across 10 dynamical systems, vs. 79, 71, and 55 for ALTRO and direct collocation (IPOPT, SNOPT); where both were valid, baselines needed $1.6\text{--}3.1\times$ the solve time.

Voice-Interactive Closed-Loop VLA Manipulation (B.E. thesis)

Feb 2026 – Aug 2026

Advisor: Prof. Yunlong Guo, Shanghai Jiao Tong University · **group leader**, team of 5

- **Deployed and fine-tuned** the $\pi_{0.5}$ VLA policy on a low-cost SO-101 arm with dual-camera observations; the team’s full system reached **70%** end-to-end success (85% pick, 75% placement) with zero safety incidents.
- **Built the VLM–policy connection**: a grounding gate that verifies spoken objects are actually in view, rewrites commands into the policy’s training-style phrasing, and blocks motion when evidence is insufficient.
- **Designed the user interface** for voice interaction and task monitoring.

HONORS & AWARDS

Honorable Mention, Interdisciplinary Contest in Modeling (ICM)

May 2023

First Prize, 13th SJTU Liming Cup Mechanical Innovation Competition

May 2023

Outstanding Volunteer, Miyuan Volunteer Team

Mar 2023

SKILLS

Methods: trajectory optimization (DDP, augmented-Lagrangian methods), VLA fine-tuning & real-robot deployment ($\pi_{0.5}$), VLM integration

Languages: Python, C++, C, MATLAB

ML & robotics: PyTorch, Transformers, OpenCV, NumPy, SciPy, OpenAI Gym, ROS, PyBullet, Gazebo

Tools: Git, Linux (Ubuntu), Docker, ~~FF~~TeX, SolidWorks